

Edith J Zhang

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EDUCATION

Columbia University, New York, NY

- Ph.D. in Applied Mathematics 2021 – 2025
Funded by NSF Graduate Research Fellowships Program

- MS in Applied Mathematics 2019 – 2021

University of Virginia, Charlottesville, VA

- BA in Mathematics 2015 – 2019

PUBLICATIONS

- [1] Edith Zhang, James Scott, Qiang Du. Graphon Limits of Graph Reaction–Diffusion Equations. *In revision at Discrete and Continuous Dynamical Systems*, 2026
- [2] Qiang Du, Kaizheng Wang, Edith Zhang, Chenyang Zhong. A particle algorithm for mean-field variational inference, *SIAM Journal on Optimization*, 2026.
- [3] Edith Zhang, James Scott, Qiang Du, Mason A. Porter, Ginzburg–Landau functionals in the large-graph limit, *Journal of Pure and Applied Functional Analysis*, 2025.
- [4] Soumyadip Ghosh, Yingdong Lu, Tomasz Nowicki, Edith Zhang, On representations of mean-field variational inference, *Journal of Applied Analysis*, 2026.
- [5] Edith Zhang, David Blei. Unveiling mode-connectivity of the ELBO landscape, *Bayesian Deep Learning Workshop*, 2021.

CONFERENCES & WORKSHOPS

- ICM, Philadelphia, PA Aug 2026
- Interacting Particle Systems, Providence, RI May 2024
Workshop at ICERM
- Bridges Conference, Richmond, VA Aug 2024
Exhibit of two mathematical artworks.
- Joint Math Meetings, Seattle, WA Jan 2024
Talk at the Complex Social Systems minisymposium titled “Higher-Dimension Opinion Dynamics”.
- Mathematics Research Communities, Java Center, NY Jun 2023
Workshop on Complex Social Systems.
- SIAM New York-New Jersey-Pennsylvania Section, Newark, NJ Oct 2023
Poster presentation titled “Ginzburg–Landau on Large Graph Limits”.
- Columbia University Data Science Day, New York, NY Apr 2022
Poster Presentation titled “VI flow: a Statistical Physics Approach to a Statistical Algorithm”.

AWARDS & SCHOLARSHIPS

- NSF Graduate Research Fellowships Program Awarded Apr 2019
- Echols Scholar at the University of Virginia Awarded Aug 2016

TEACHING AND OUTREACH

Instructor, University of California, Los Angeles (UCLA)

- Math 134: Linear and Nonlinear Systems of Dynamical Systems (Two sections) Fall 2026
- Math 135: Ordinary Differential Equations Spring 2026
- Math 134: Linear and Nonlinear Systems of Dynamical Systems Winter 2026
- Math 31A: Calculus I (Two sections) Fall 2025

Adjunct Instructor, The Cooper Union, New York, NY

- Calculus II, Spring 2025

Teaching Assistant, Columbia University, New York, NY

- Partial Differential Equations Fall 2019
- Mathematics for data science Spring 2020
- Linear algebra Fall 2021
- Intro to numerical methods Spring 2022
- Linear algebra Fall 2022
- Introduction to Numerical Methods Fall 2023

Applied Mathematics Graduate Student Seminar, Columbia University, New York, NY 2022-2024

- Initiated and co-lead weekly seminar for graduate students to present topics relating to their research, hard and soft skills, job search, writing, etc.

TALKS

Graphon Reaction–Diffusion Equations

- Talk at Networks Journal Club, UCLA
- Talk at Level Set Seminar, UCLA

January 2026
February 2026

Higher-Dimension Opinion Dynamics

- Talk at Complex Social Systems minisymposium at the Joint Math Meetings

January 2024

Two mathematical artworks

- Exhibition at the Bridges conference on mathematical connections in art, music, architecture, and culture

August, 2024

Ginzburg–Landau on Large Graph Limits

- Poster presentation at SIAM-NNP Conference, New Jersey Institute of Technology, Newark, NJ

Oct 2023

VI flow: a statistical physics approach to a statistical algorithm

- Data Science Day, Columbia University, New York, NY

April 2022

Methods of numerical and optimization in mathematical modeling

- Undergraduate Math Modeling Workshop, Columbia University, New York, NY

Feb 2021

Unveiling mode-connectivity of an optimization landscape

- APAM research conference, Columbia University, New York, NY
- Data Science Day, Columbia University, New York, NY

Mar 2021

April 2021